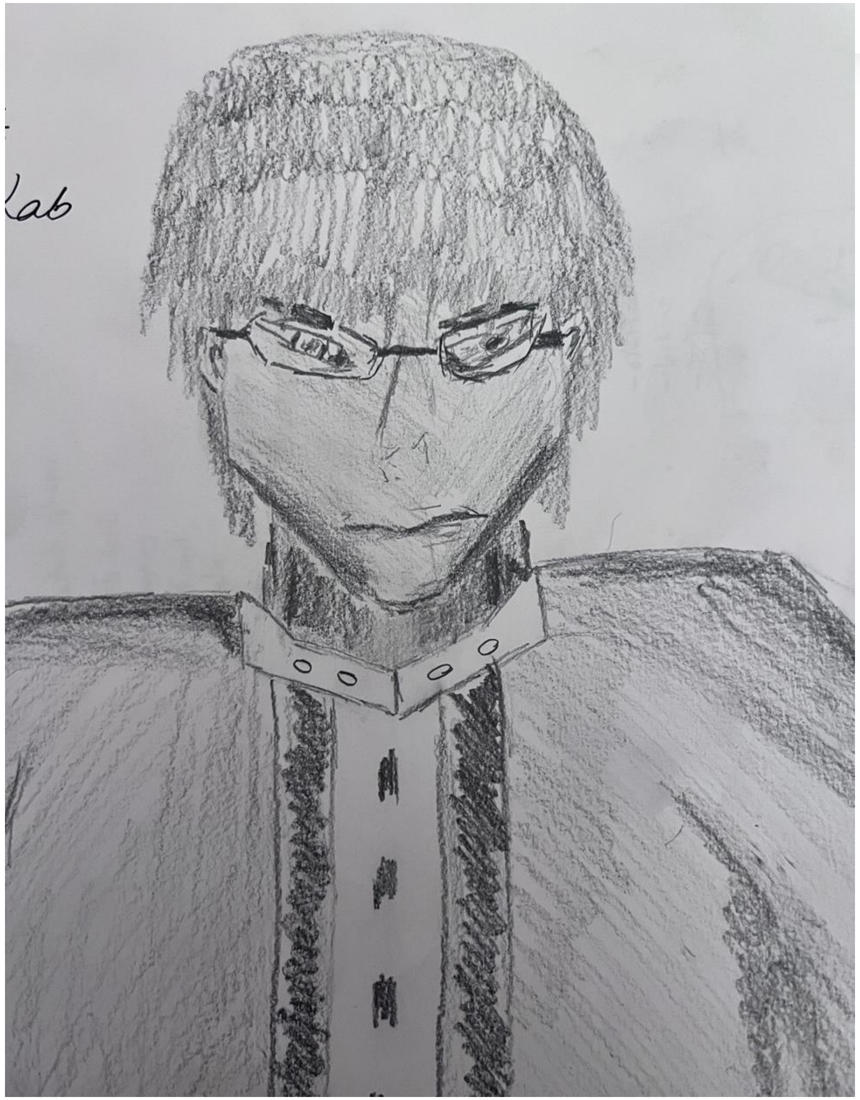
Importance of Diversity & Representation in Science

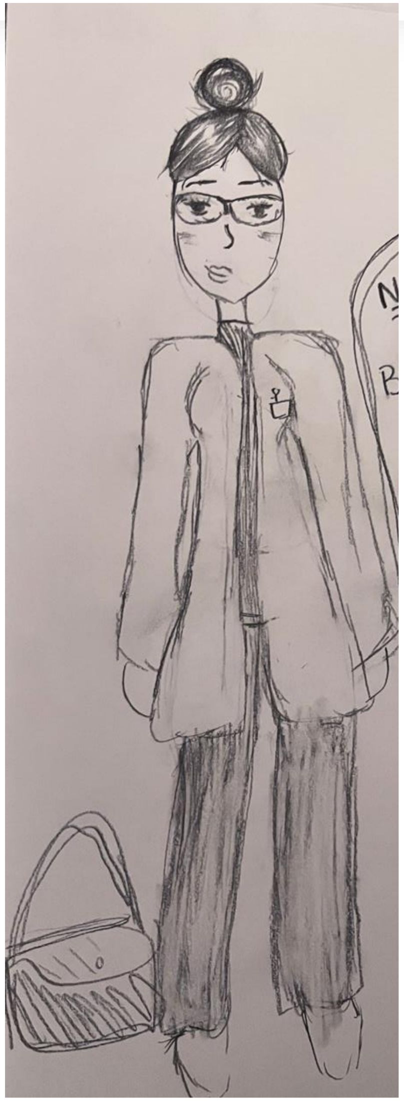
Image: ©Mykola – stock.adobe.com

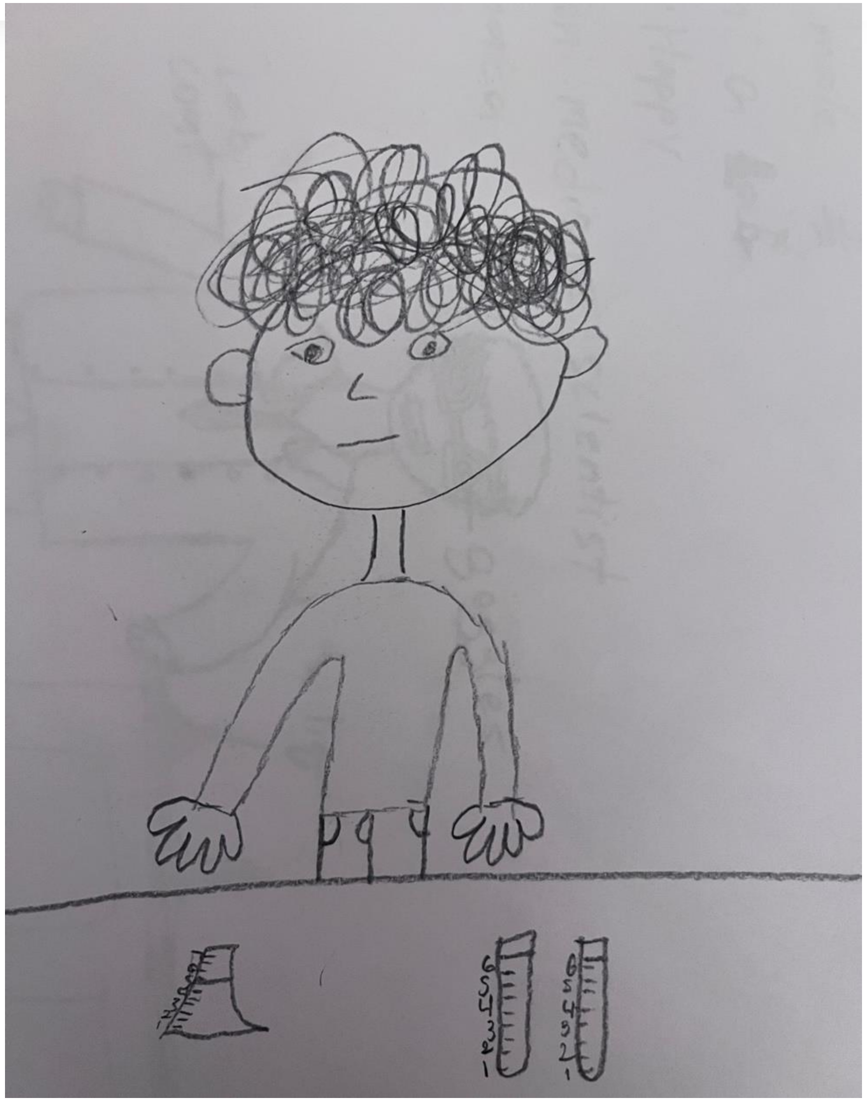
What does a scientist look like?

What does a scientist do?

What does a scientist look like?







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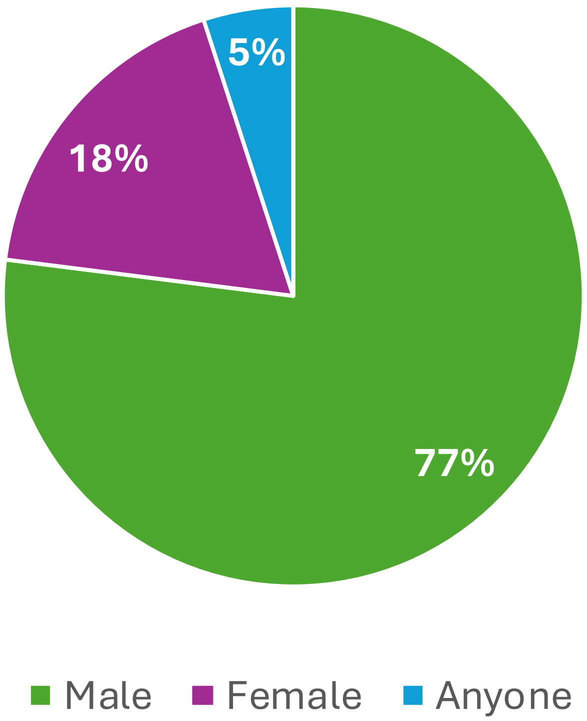

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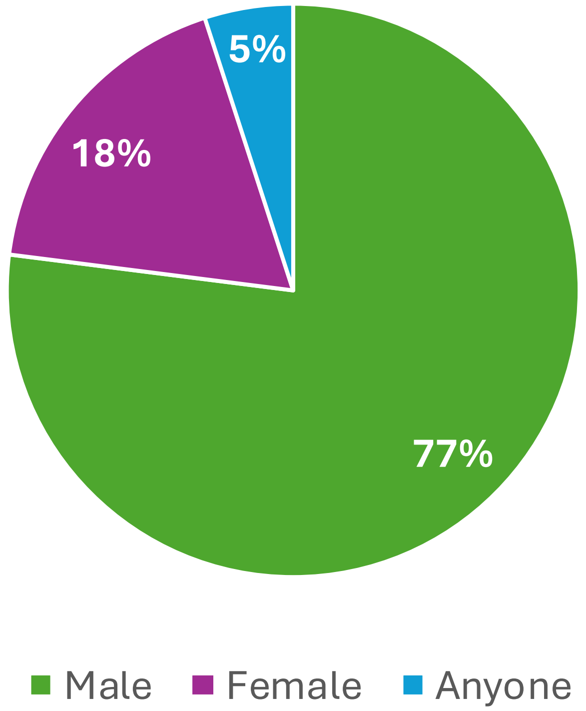

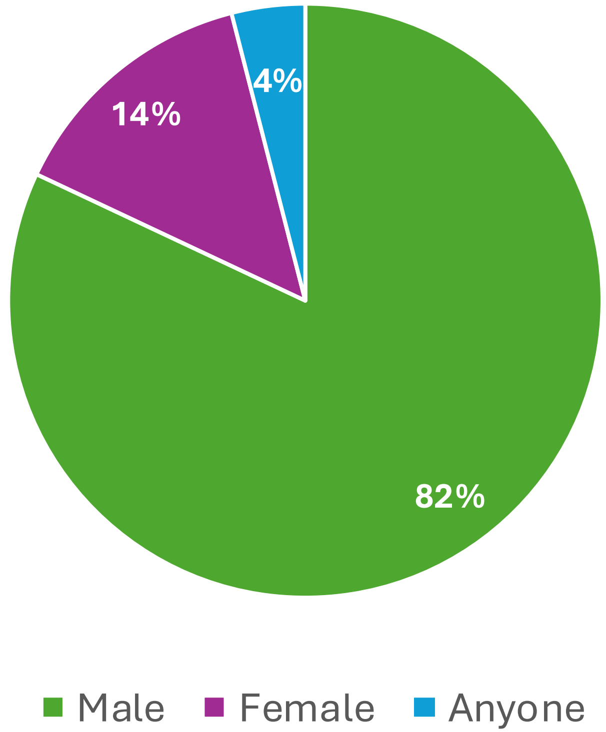

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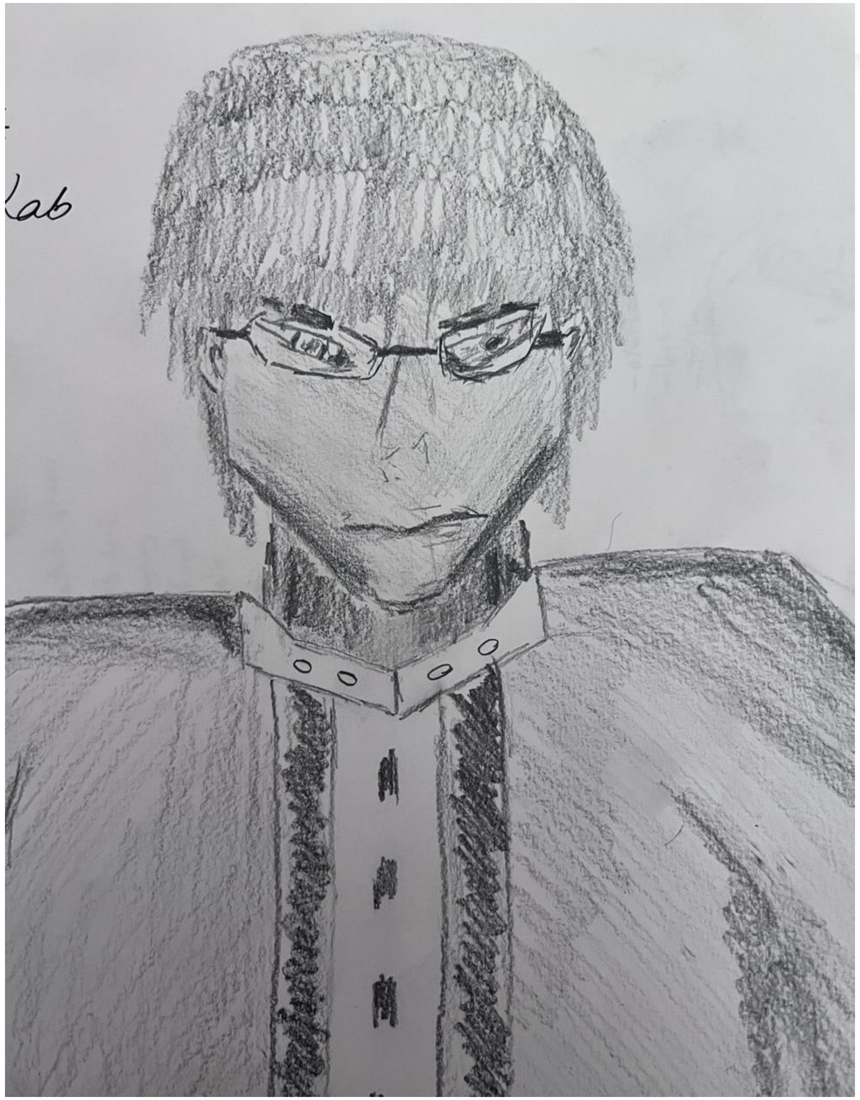

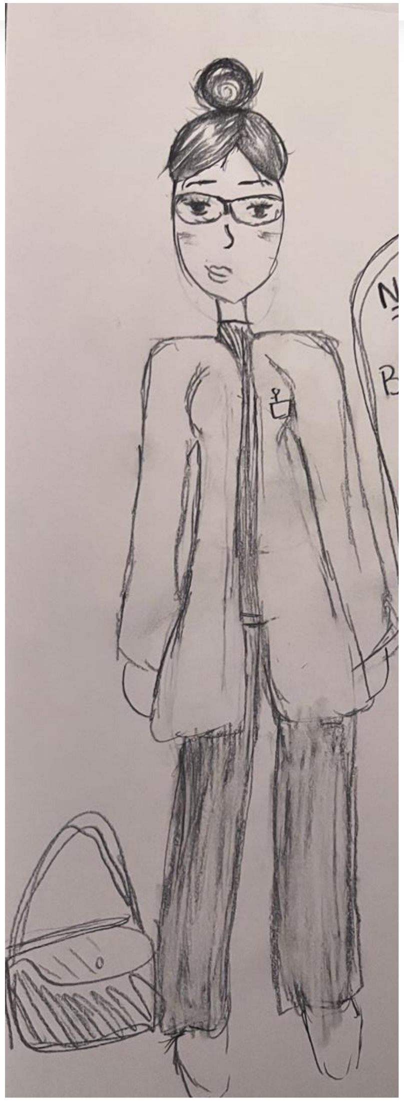

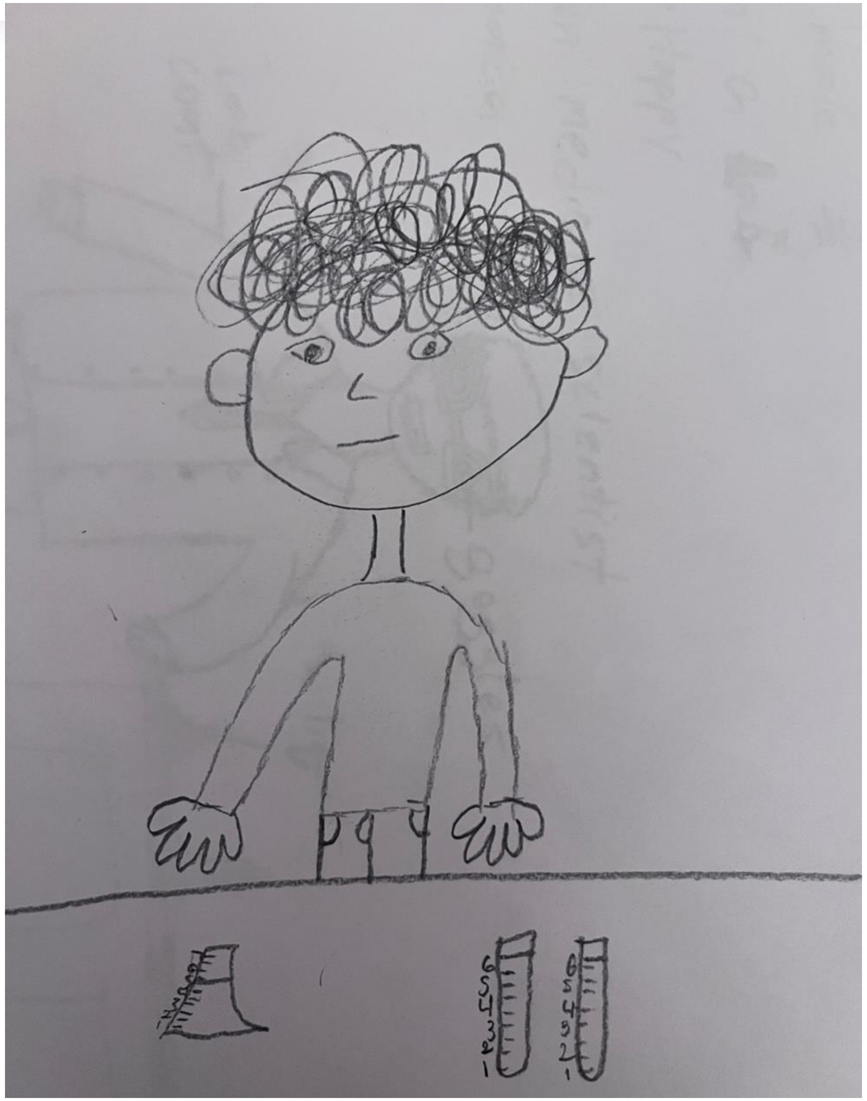

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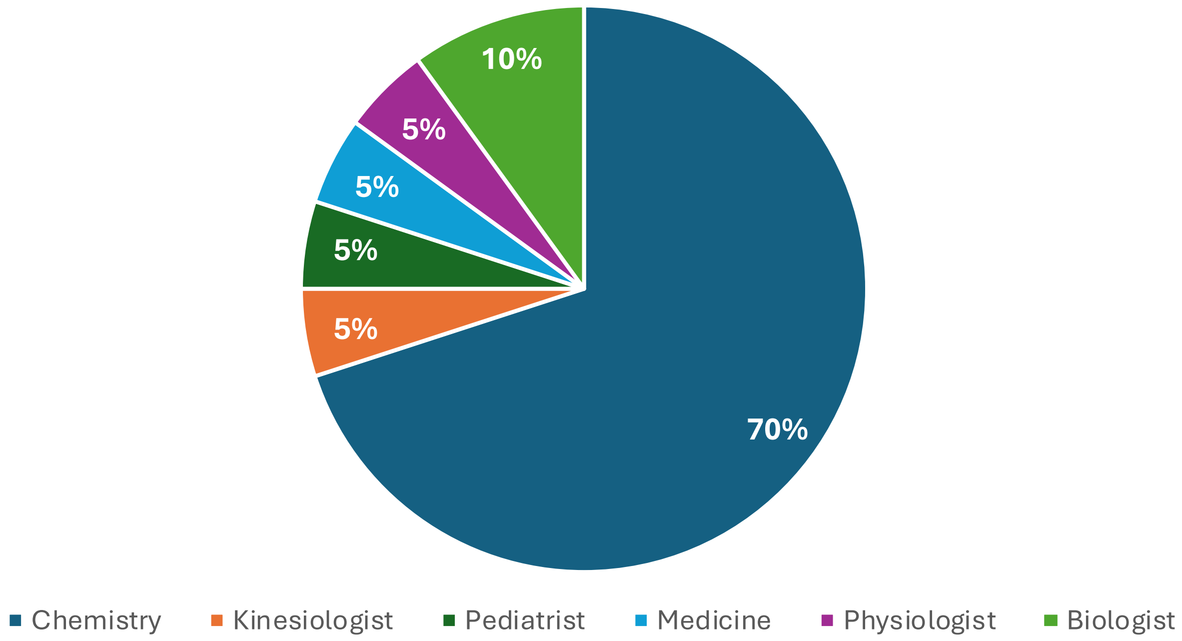

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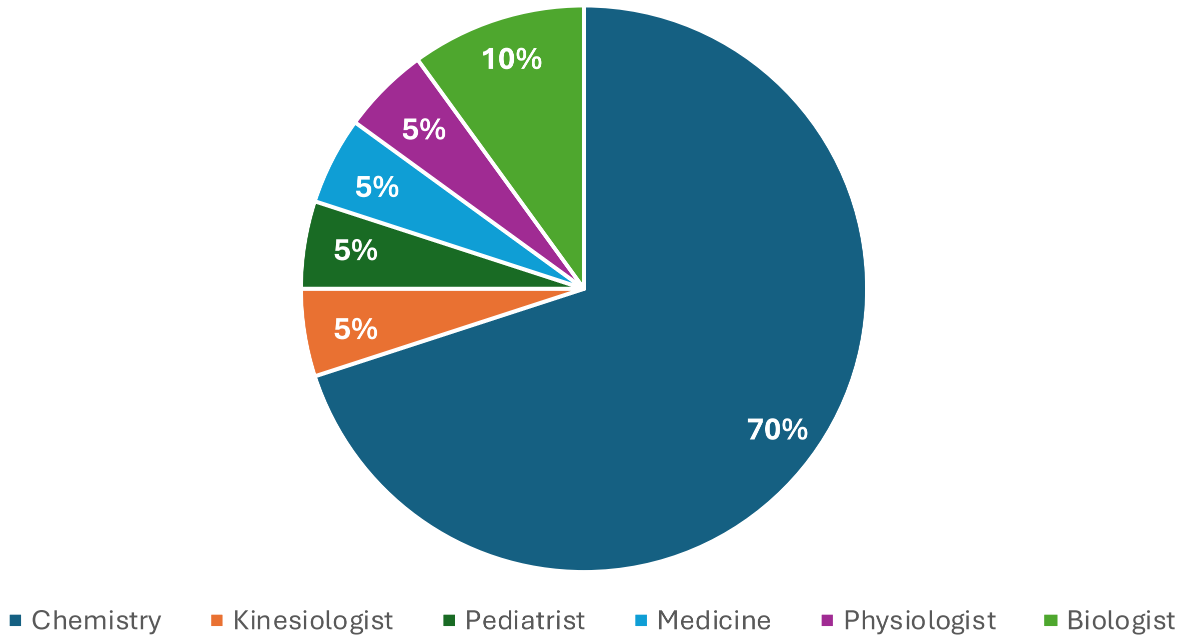
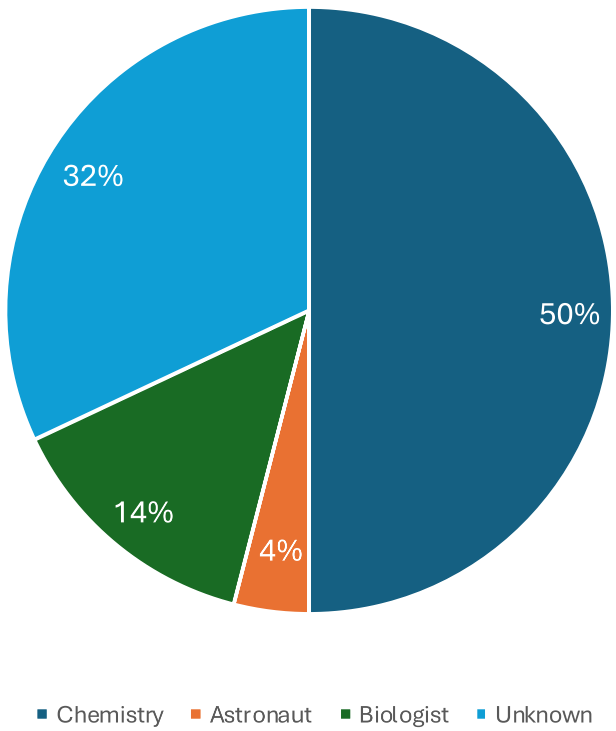


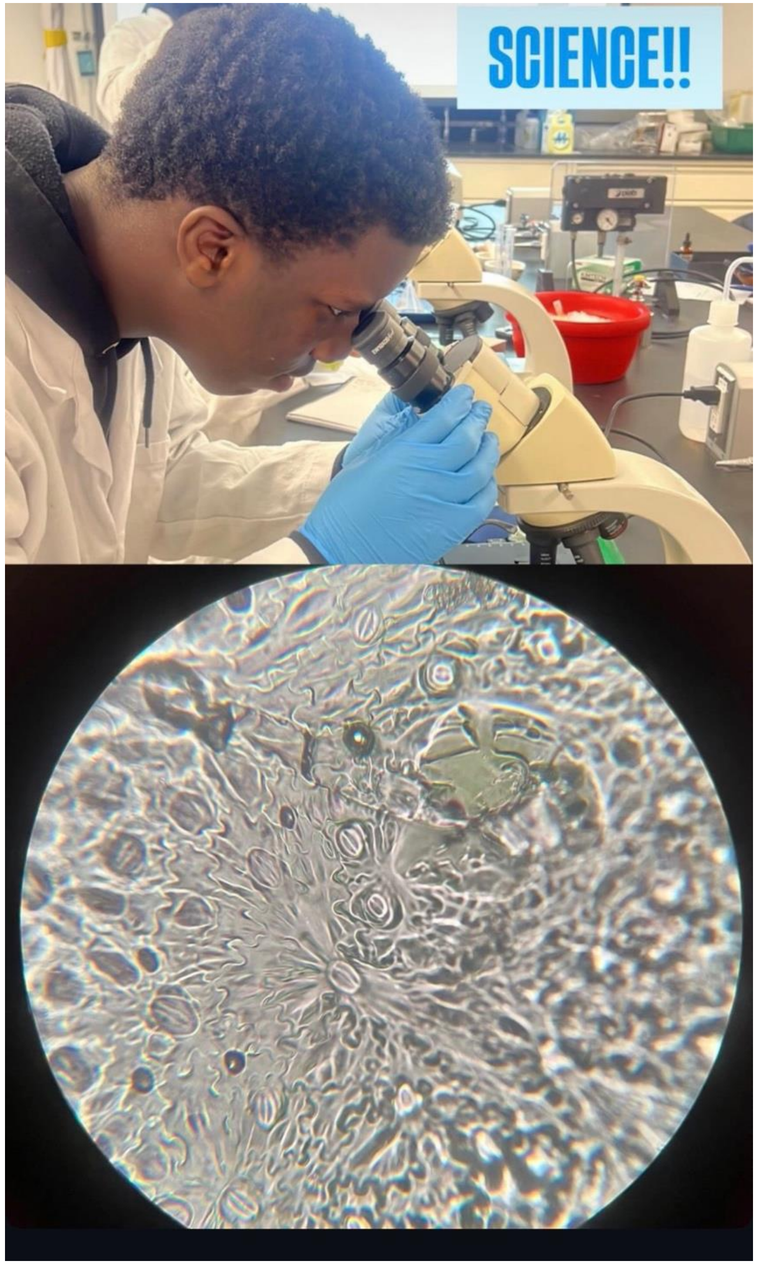


What do you think about the individuals in these pictures?

What do you think they all have in common?

What do you think about the individuals in these pictures?

What do you think they have in common with the previous pictures ?



Why we think that all scientist mostly do the same thing? And look like Chris?

Where is that idea coming from?

Why we think that all scientist mostly do the same thing? And look like Chris?

Where is that idea coming from?

What is Bias?

Bias

- Any thought or way of thinking that can make you like or dislike someone or something without evidence

  A computer science teacher thinks only students with good math grade will be good coders,

  so they only invite students with good math grades to join the coding club

- The fact of preferring a particular subject or thing

  Luciana showed a biology bias since child

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Are you always aware of your biases?

Bias

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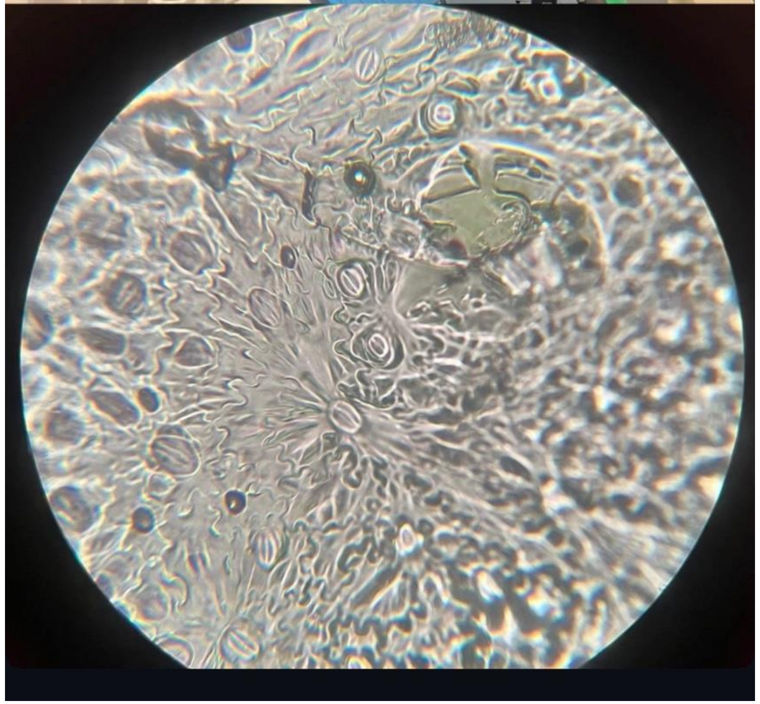
Unconscious Bias

- Any thought or way of thinking that can make you like or dislike something or someone **without you even noticing**

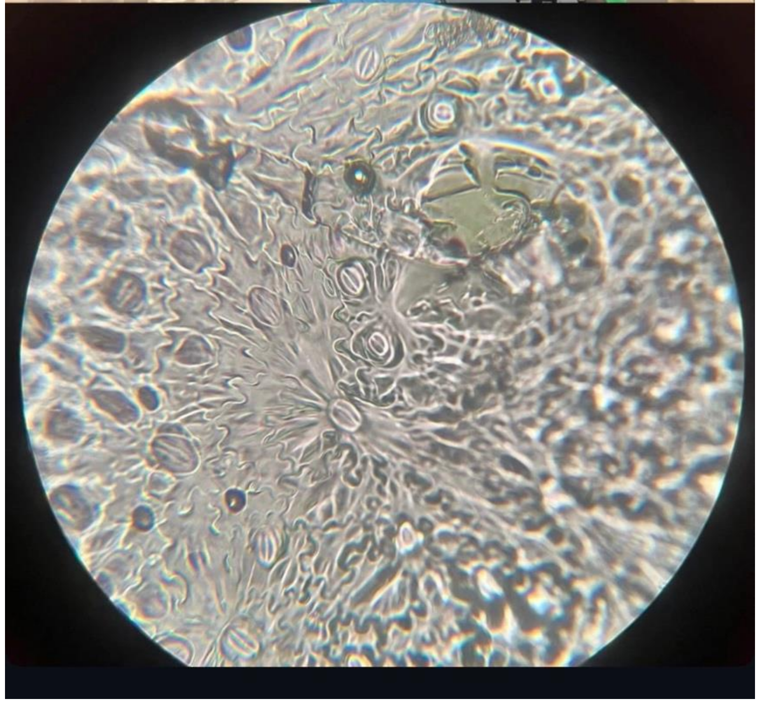
The computer science teacher watches a lot of movies where the hackers always wear glasses, they then ***happen to*** *only invite students who wear glasses to join the coding club*

Maybe we should define what a scientist is?

Anyone can be a scientist



Everyone
~~Anyone~~ can be a scientist



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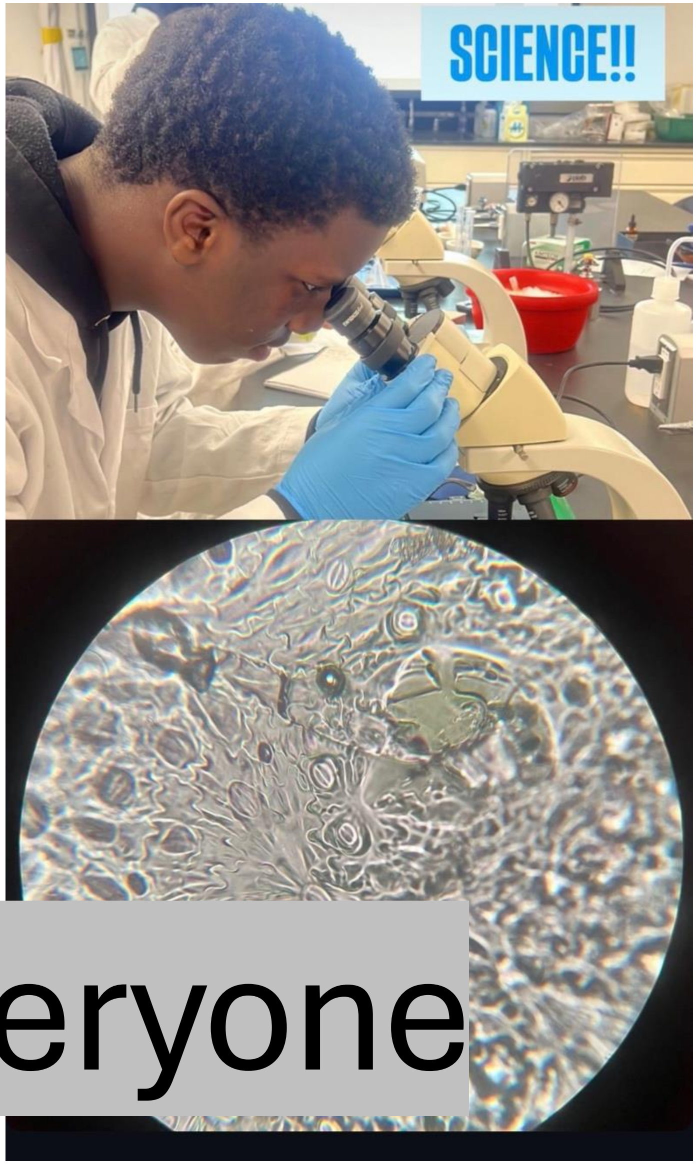


Science Needs Everyone

	True	False	Not sure
Science is always correct			
Scientific knowledge is always tentative and subject to change			
Scientific results are based on evidence gathered through observation, experimentation and theory			
Scientists validate their findings to a peer review process			
Scientists are trying to answer questions to understand the world			
Scientists do not change their minds			
Scientists are often very serious people			
Scientists do not involve any emotions, or politics in their research			

Scientists are expected to be
(and often think they are)
impartial, apolitical,
and without bias.
But they often are not.

Working in Groups

Females wearing seat belts are 47% more likely than male seat belt-wearers to be severely injured

The Solution: Female Crash Test Dummies

ELSEVIER

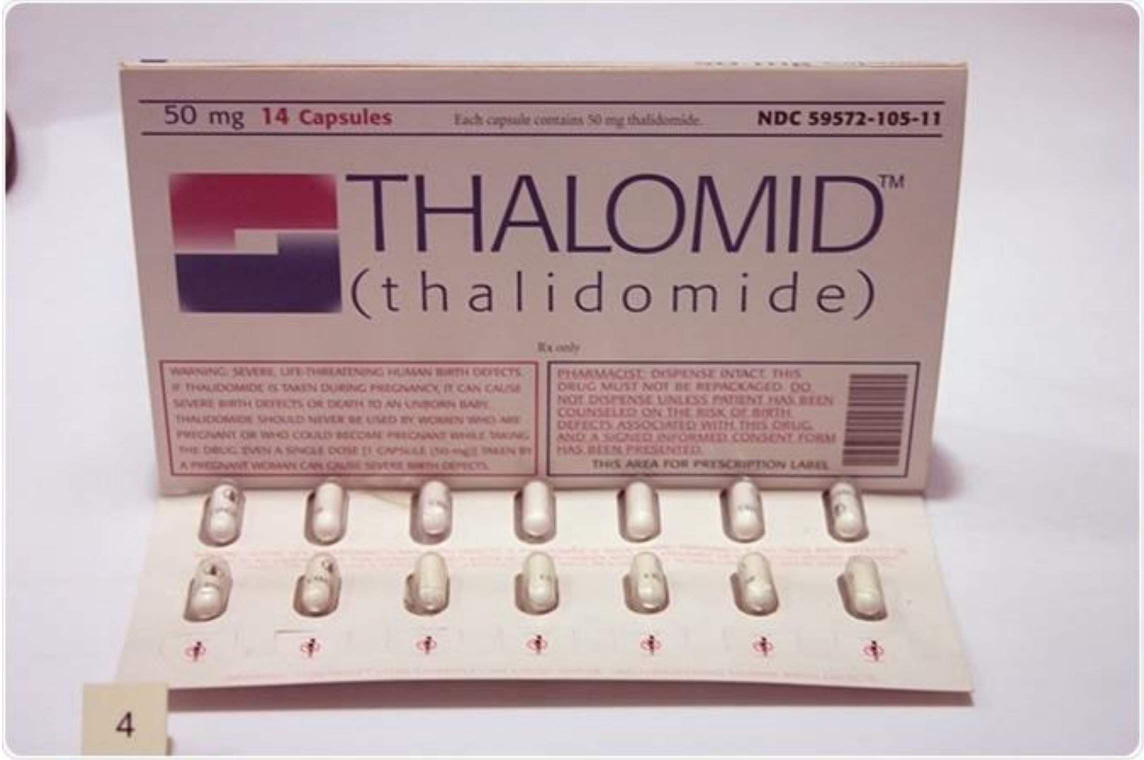
Accident Analysis & Prevention

Volume 127, June 2019, Pages 156-162

Review of average sized male and female occupant models in European regulatory safety assessment tests and European laws: Gaps and bridging suggestions

Astrid Linder [a b], Wanna Svedberg [a]

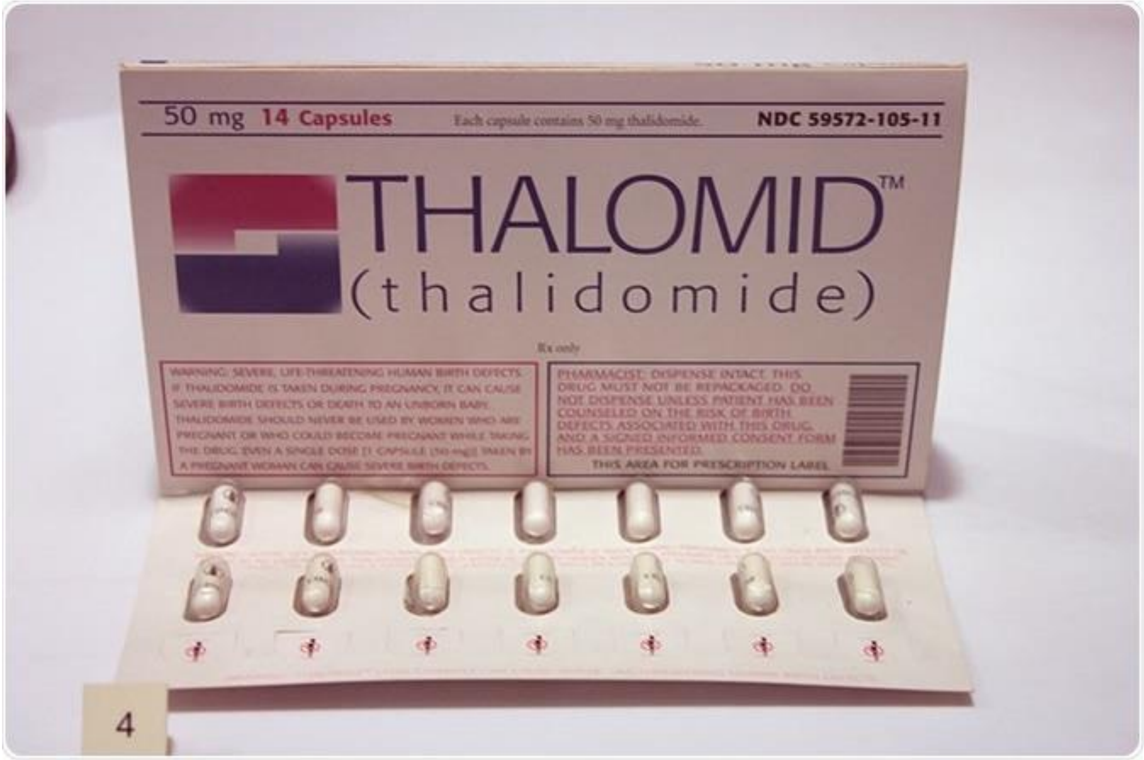
The Thalidomide Crisis



Drug developed as a sedative and was marketed for various conditions

Morning sickens in pregnant woman

The Thalidomide Crisis



Drug developed as a sedative and was marketed for various conditions

Morning sickens in pregnant woman

Resulted in over 10.000 of children's born with sever malformations

AI Bias in Resume Selection

Gender, Race, and Intersectional Bias in Resume Screening via Language Model Retrieval

Kyra Wilson, Aylin Caliskan

University of Washington Information School
Seattle, Washington, USA
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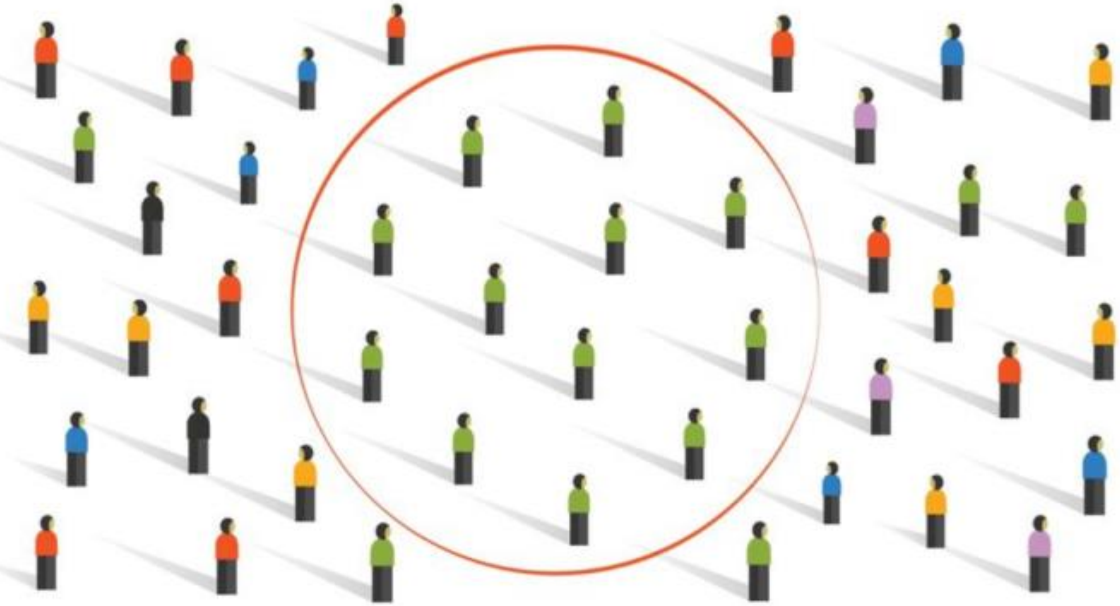
ChatGPT



AI Bias Impact Hiring and Recruitment

GO PLATFORM
Education | Recruitment | Development

What is the common issue in these examples?

Working in Groups

Monoculture vs Polyculture

Breaking down the biases

Image: ©Mykola – stock.adobe.com

Diversity And Inclusion

Why Diverse Teams Are Smarter

by David Rock and Heidi Grant

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Process info more carefully

Focus more on facts

Are more innovative

Diverse perspectives, life experiences, identities and backgrounds

=

Better Science!